

# MUTUAL FUND ANALYTICS PLATFORM

Final Project Report

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# 1. Executive Summary

## 1.1 Business Overview

**Mutual Fund Analytics Platform** is an end-to-end data analytics solution developed to transform fragmented mutual fund data into meaningful business insights. The project integrates historical NAV data, investor transactions, SIP inflows, AUM statistics, and benchmark information using Python, SQL, and Power BI. By consolidating data from multiple public sources into a centralized analytical platform, the solution enables efficient fund performance evaluation, investor behavior analysis, and interactive reporting for informed decision-making.

## 1.2 Objectives

- Develop an automated ETL pipeline to extract, clean, transform, and load mutual fund data.
- Design a SQL-based star schema for efficient data storage and querying.
- Perform Exploratory Data Analysis (EDA) to identify investment trends and investor behavior.
- Compute key financial performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, and Beta.
- Build an interactive Power BI dashboard to visualize fund performance, investor analytics, and market trends.

## 1.3 Outcome

The project successfully delivered a centralized mutual fund analytics platform capable of processing large financial datasets and generating actionable insights. The ETL pipeline improved data quality and consistency, while analytical models enabled risk-adjusted fund comparison. The Power BI dashboard provides an interactive view of industry trends, investor demographics, transaction patterns, and fund performance, supporting data-driven investment decisions.

## 1.4 Key Performance Indicators (KPIs)

- **40** Mutual Fund Schemes Analyzed
- **87K+** Historical NAV Records Processed
- **32K+** Investor Transactions Analyzed
- **4-Page** Interactive Power BI Dashboard Developed
- **Star Schema** SQL Database Implemented
- **6+** Risk & Performance Metrics Computed (CAGR, Sharpe, Sortino, Alpha, Beta, Max Drawdown)

## 2. Data Sources

### 2.1 Data Sourcing

The Mutual Fund Analytics Platform integrates data from multiple publicly available financial sources to provide a comprehensive view of mutual fund performance and investor activity. A hybrid data collection approach was adopted, combining both static datasets and live API data.

- **Static Data:** The primary datasets were obtained from AMFI India and included ten structured CSV files containing fund master information, historical NAV records, Assets Under Management (AUM), SIP inflows, category-wise fund flows, investor transactions, portfolio holdings, and benchmark indices.
- **Dynamic Data:** To complement the historical datasets, a Python-based data ingestion script was developed to retrieve the latest Net Asset Value (NAV) data from the mfapi.in REST API. This enabled periodic updates for selected mutual fund schemes, ensuring that the analytics platform reflects recent market information.

### 3. Dataset Description

The project combines multiple datasets covering different aspects of the mutual fund ecosystem, including:

- **Fund Master:** Contains scheme information such as fund house, category, expense ratio, benchmark, and risk level.
- **Historical NAV Data:** Daily NAV values for multiple mutual fund schemes used to calculate returns and risk metrics.
- **Assets Under Management (AUM):** Quarterly AUM information for major Asset Management Companies (AMCs).
- **Monthly SIP Inflows:** Monthly SIP investment amounts, active SIP accounts, and year-over-year growth statistics.
- **Category-wise Net Inflows:** Monthly inflow data across various mutual fund categories such as Large Cap, Mid Cap, Small Cap, Hybrid, ELSS, and Liquid Funds.
- **Investor Transactions:** Transaction records including SIP, Lumpsum, and Redemption investments along with investor demographics such as age group, gender, state, and city tier.
- **Benchmark Indices:** Historical benchmark values used for evaluating fund performance relative to the broader market.
- **Portfolio Holdings:** Sector-wise allocation of equity mutual fund portfolios used for concentration and diversification analysis.

#### 3.1 Data Characteristics

- **Historical Coverage:** January 2022 – May 2026
- **Mutual Fund Schemes:** 40
- **Historical NAV Records:** Approximately 87,000+
- **Investor Transactions:** Approximately 32,000+
- **Asset Management Companies (AMCs):** 10
- **Data Formats:** CSV, JSON (REST API), and SQL Tables

The collected datasets combination of historical and live data provide a reliable foundation for performing financial performance analysis, risk assessment, investor segmentation, market trend analysis, and interactive dashboard development while maintaining data consistency and integrity throughout the analytics pipeline.

## 4. ETL Pipeline

The ETL (Extract, Transform, Load) pipeline forms the backbone of the Mutual Fund Analytics Platform by ensuring that data from multiple sources is collected, cleaned, transformed, and stored in a structured format for analysis. The pipeline was developed using Python, Pandas, SQLAlchemy, and SQLite, enabling an automated and reliable data processing workflow.

### 4.1 Extract

The extraction phase involved collecting data from multiple sources, including AMFI India datasets, historical NAV records, investor transaction files, SIP inflow reports, benchmark indices, and portfolio holdings. Additionally, live Net Asset Value (NAV) data was retrieved using the mfapi.in REST API to keep selected mutual fund schemes up to date. The extracted data was stored in CSV format for further processing.

### 4.2 Transform

During the transformation stage, the raw datasets were cleaned and standardized to improve data quality and consistency. The major transformation steps included:

- Handling missing values using appropriate imputation techniques, including forward filling (ffill) for missing NAV values on non-trading days.
- Removing duplicate records to eliminate data redundancy.
- Standardizing date formats and converting data types to ensure compatibility across datasets.
- Validating fund identifiers (AMFI Codes) and maintaining referential integrity between related datasets.
- Standardizing categorical fields such as transaction type, fund category, and investor demographics.
- Creating derived attributes such as daily returns, monthly returns, and year-over-year growth to support analytical reporting.
- Performing data validation checks to ensure numerical fields contained valid values and records met predefined quality criteria.

These transformations produced a clean, analysis-ready dataset suitable for SQL querying and visualization.

#### 4.2.1 Data Transform Results

The raw datasets were systematically cleaned and validated to ensure high data quality before analysis and visualization. The cleaning process involved handling missing values, removing duplicate records, validating financial data, and standardizing categorical fields.

| Dataset               | Cleaning Results                                                                                                                                                                                                                                                      |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NAV History           | Processed <b>46,000</b> raw records into <b>64,320</b> cleaned records after adding <b>18,320</b> weekend and market holiday entries using scheme-wise forward-fill (ffill). The final dataset contained <b>0 duplicate records</b> and <b>0 invalid NAV values</b> . |
| Investor Transactions | Processed <b>32,778</b> transaction records with <b>0 duplicate entries</b> , <b>0 missing values</b> , and <b>0 invalid</b> transaction amounts, transaction types, or KYC statuses after data standardization and validation.                                       |

The data cleaning process significantly improved dataset completeness, consistency, and reliability. The cleaned datasets served as the foundation for subsequent SQL analysis, financial performance calculations, exploratory data analysis (EDA), and Power BI dashboard development.

#### 4.3 Load

The transformed datasets were loaded into a SQLite relational database using SQLAlchemy. A Star Schema data model was implemented to improve query performance and simplify analytical reporting. Dimension tables, including Fund, Date, and Investor, were linked to fact tables containing NAV history, transaction records, and fund performance metrics through primary and foreign key relationships.

The centralized database served as the primary data source for SQL analysis and the Power BI dashboard, enabling efficient querying, aggregation, and visualization of financial and investor data.

#### 4.4 ETL Workflow

The overall ETL workflow followed these sequential steps:

1. Extract data from CSV files and live APIs.
2. Validate and clean the raw datasets.
3. Standardize formats and perform feature engineering.
4. Load the processed data into the SQLite database.
5. Execute SQL queries for analytical reporting.
6. Connect Power BI to the database for interactive dashboard creation.

This automated ETL pipeline ensured high data quality, minimized manual intervention, and provided a scalable foundation for mutual fund analytics and business intelligence.

## 5. Database Schema

To support efficient storage, querying, and analytical reporting, the Mutual Fund Analytics Platform uses a Star Schema data model implemented in SQLite. This schema organizes data into fact tables containing measurable business events and dimension tables containing descriptive attributes. The design minimizes data redundancy while improving query performance for SQL analysis and Power BI visualizations.

### 5.1 Schema Design

The database consists of three primary fact tables connected to multiple dimension tables through primary and foreign key relationships.

#### *Dimension Tables*

- **dim\_fund** – Stores master information for each mutual fund scheme, including AMFI code, fund house, category, benchmark, expense ratio, and risk level.
- **dim\_date** – Contains calendar attributes such as date, month, quarter, and year, enabling time-based analysis.

#### *Fact Tables*

- **fact\_nav** – Stores historical Net Asset Value (NAV) records for each mutual fund scheme along with the corresponding date.
- **fact\_transactions** – Records investor transactions such as SIP, Lumpsum, and Redemption amounts linked to investor and fund information.
- **fact\_performance** – Contains calculated financial performance metrics, including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Standard Deviation, and Maximum Drawdown for each mutual fund scheme.
- **fact\_investor** – Stores investor demographic details, including age group, gender, state, city tier, annual income, and KYC status.

### 5.2 Entity Relationships

The AMFI Code serves as the primary key for linking fund-related information across the database. Similarly, Date IDs and Investor IDs establish relationships between transactional and descriptive data. This structure enables efficient aggregation of financial metrics across different dimensions such as fund house, category, investor demographics, and time periods.

### 5.3 Benefits of the Star Schema

The Star Schema was selected because it offers several advantages for analytical applications:

- Reduces data redundancy through dimensional modeling.
- Improves SQL query performance for aggregation and reporting.

- Simplifies integration with Power BI for interactive dashboards.
- Supports multidimensional analysis across funds, investors, and time.
- Provides a scalable foundation for future enhancements and additional datasets.

The implemented database schema serves as the central repository for all processed data and enables efficient execution of SQL queries, financial performance calculations, and business intelligence reporting throughout the Mutual Fund Analytics Platform.

## 6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to uncover key trends in the mutual fund industry, investor behavior, and market performance. The analysis provided valuable insights that guided the subsequent performance evaluation and dashboard development.

### 6.1 EDA Findings

- **Industry Growth:** The Indian mutual fund industry experienced significant expansion, with total Assets Under Management (AUM) reaching approximately ₹81 Lakh Crore, reflecting sustained investor confidence and market growth.
- **SIP Growth:** Monthly Systematic Investment Plan (SIP) inflows recorded a steady upward trend, reaching an all-time high of ₹31,002 Crore, indicating increasing adoption of disciplined, long-term investing.
- **Fund House Analysis:** Leading Asset Management Companies (AMCs), particularly SBI Mutual Fund, dominated the industry in terms of Assets Under Management, highlighting strong brand presence and investor trust.
- **Investor Demographics:** Investors aged 26–35 years formed the largest investment segment, while Tier-1 cities and states such as Maharashtra and Karnataka contributed the highest transaction volumes, highlighting growing opportunity in B30 (Beyond Top 30) cities.
- **Transaction Behaviour:** SIP investments accounted for the majority of investor transactions, demonstrating a clear preference for systematic wealth creation over lump-sum investing.
- **Portfolio Composition:** Equity mutual funds represented the largest share of analyzed schemes, with significant allocations towards the Financial Services and Information Technology sectors, emphasizing the importance of portfolio diversification.

### 6.2 Visualizations

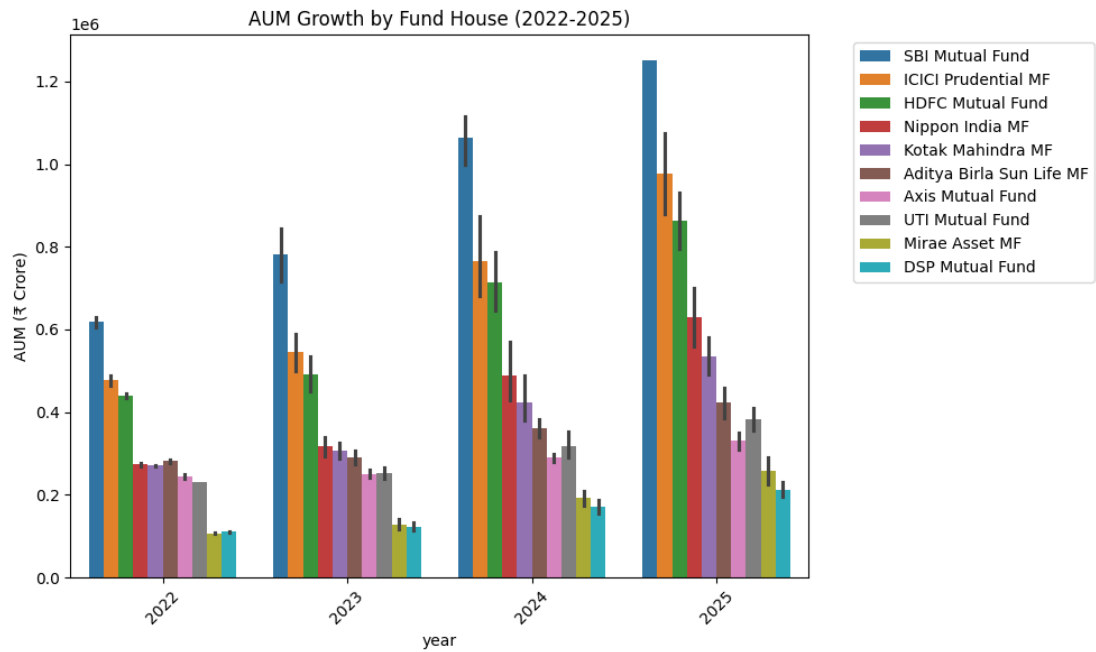


Figure 6.1: Industry AUM Growth and Fund House Comparison

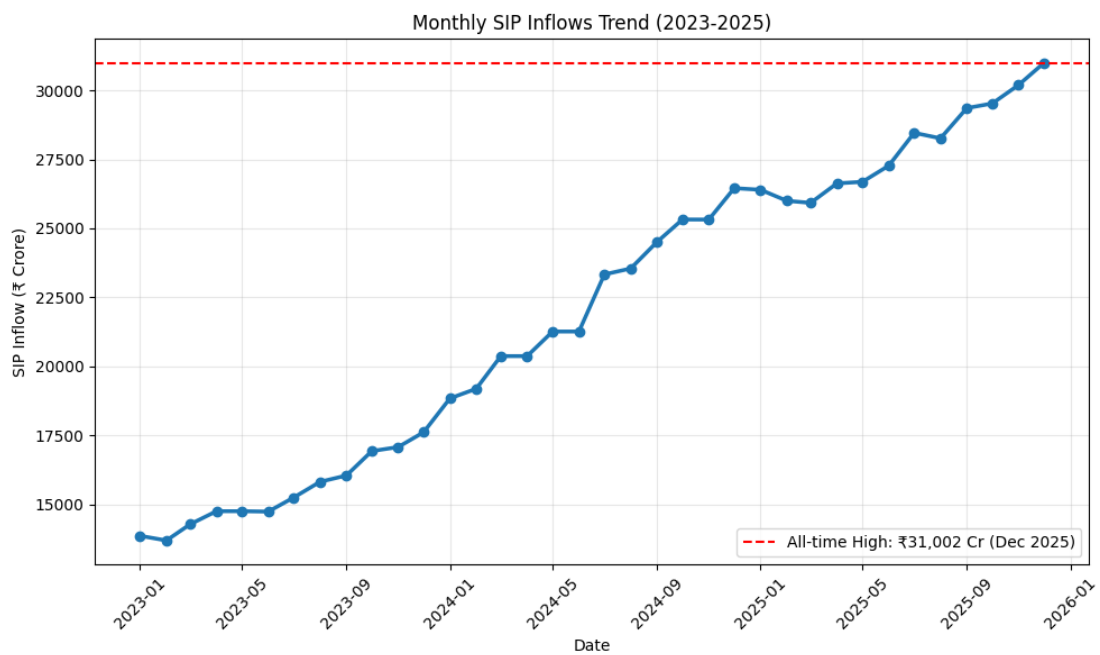


Figure 6.2: Monthly SIP Inflow Trend

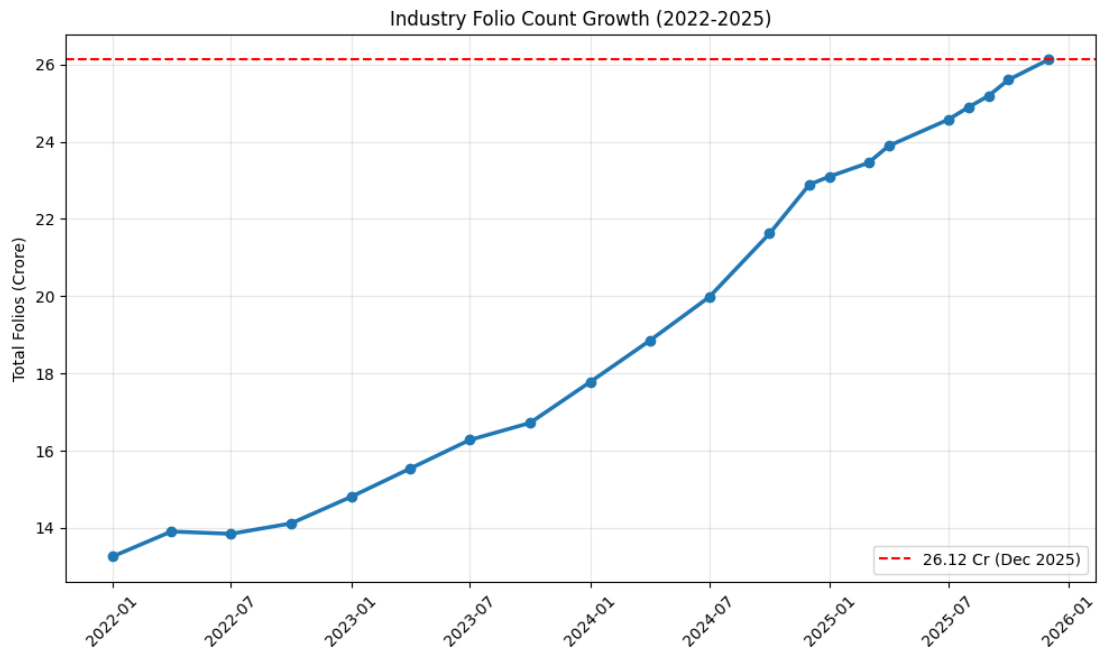


Figure 6.3: Growth of Active Mutual Fund Folios

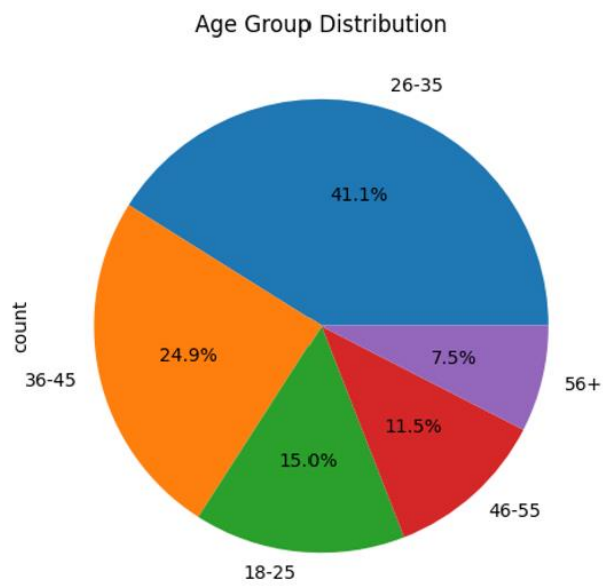


Figure 6.4: Investor Age Group Distribution

## 8. Investor Gender Split

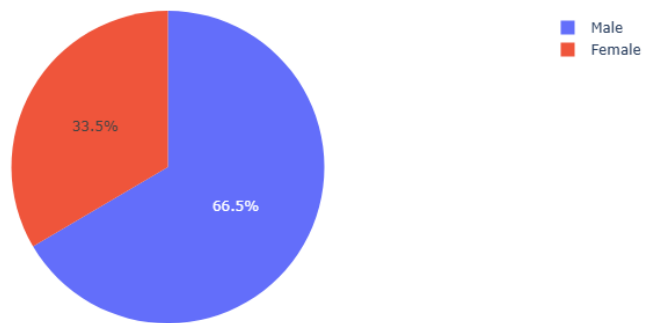


Figure 6.5: Gender-wise Investor Distribution

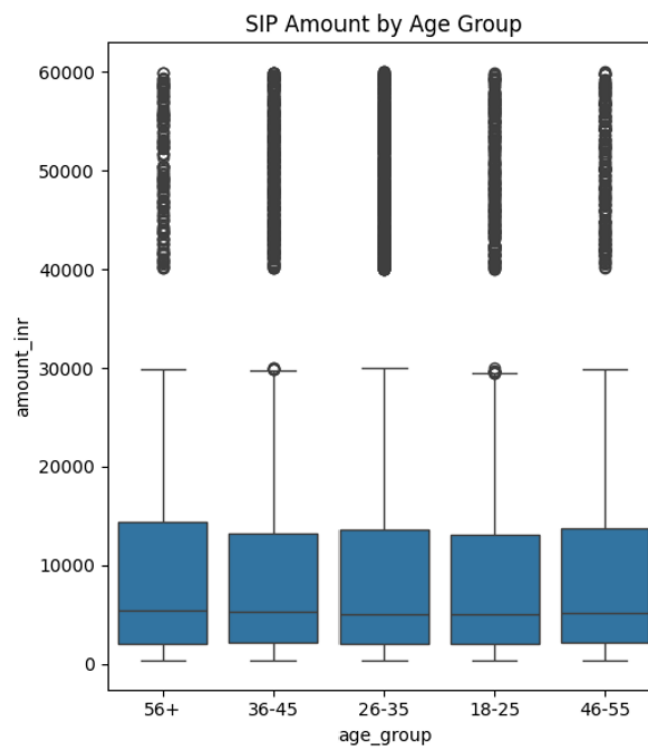


Figure 6.6: SIP Allocation Across Age Groups

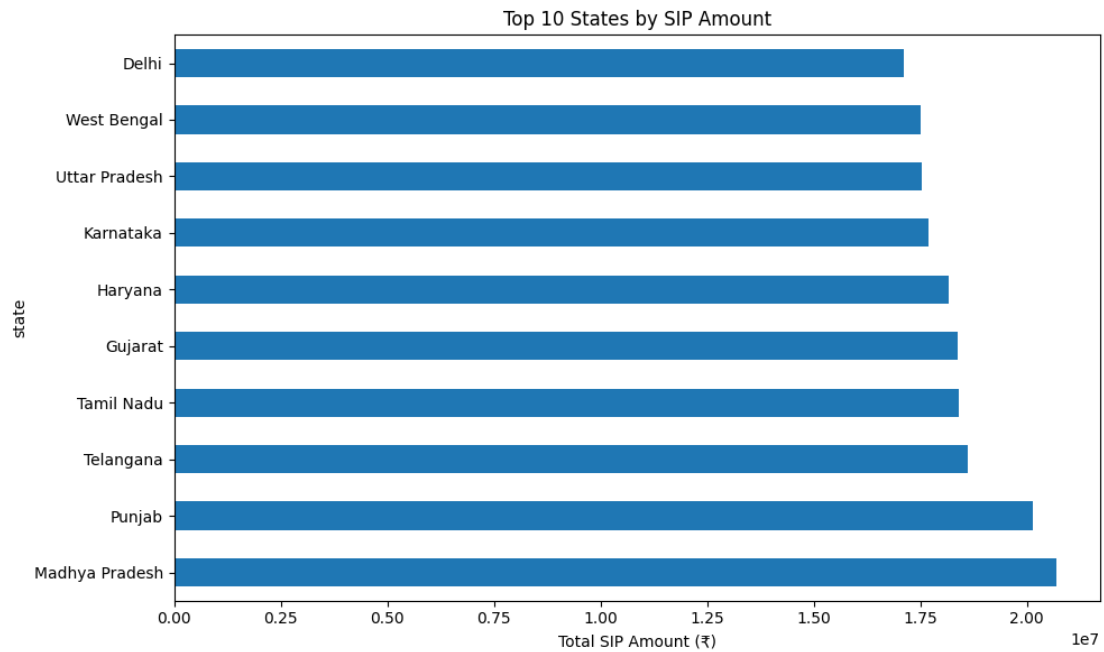


Figure 6.7: State-wise SIP Investment Distribution

## 11. city Tier Distribution (T30: Top 30 Cities, B30:Beyond 30)

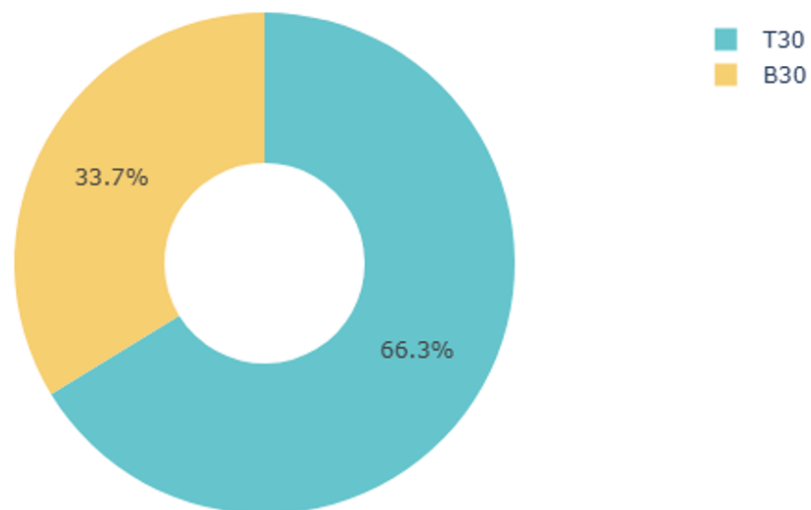


Figure 6.8: Market Penetration by City Tier

The EDA findings reveal a rapidly growing mutual fund ecosystem characterized by increasing retail participation, strong SIP adoption, and concentration of investments in major metropolitan regions. These insights provide the foundation for evaluating fund performance, assessing investment risk, and developing strategic business recommendations.

## 7. Fund Performance & Advanced Risk Analysis

Moving beyond simple point-to-point returns, the project evaluated funds on institutional-grade risk-adjusted metrics.

### 7.1 Quantitative Performance Metrics

- **CAGR & Volatility:** Calculated 1Y, 3Y, and 5Y Compound Annual Growth Rates.
- **Risk-Adjusted Returns:** Computed the Sharpe Ratio (assuming a 6.5% risk-free rate) and Sortino Ratio (measuring downside deviation).
- **Market Regression:** Executed OLS regression against the Nifty 50 benchmark to extract annualized Alpha (excess return) and Beta (market correlation).

These metrics were aggregated into a proprietary **Custom Fund Scorecard (0–100)** to objectively rank the top 40 schemes.

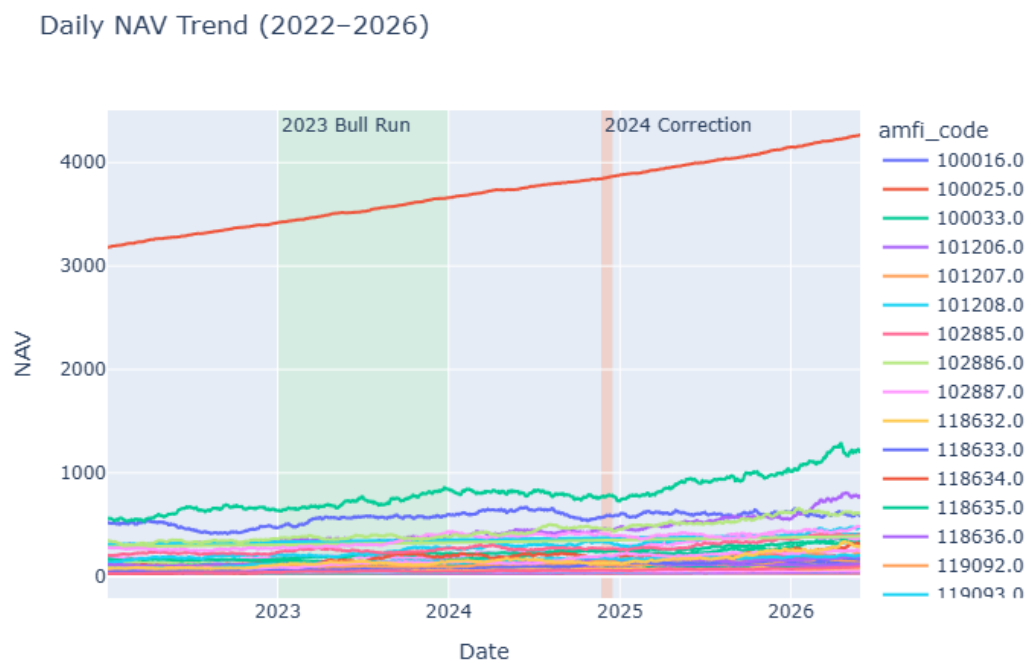


Figure 7.1: Historical NAV Trajectory for Evaluated Mutual Fund Schemes

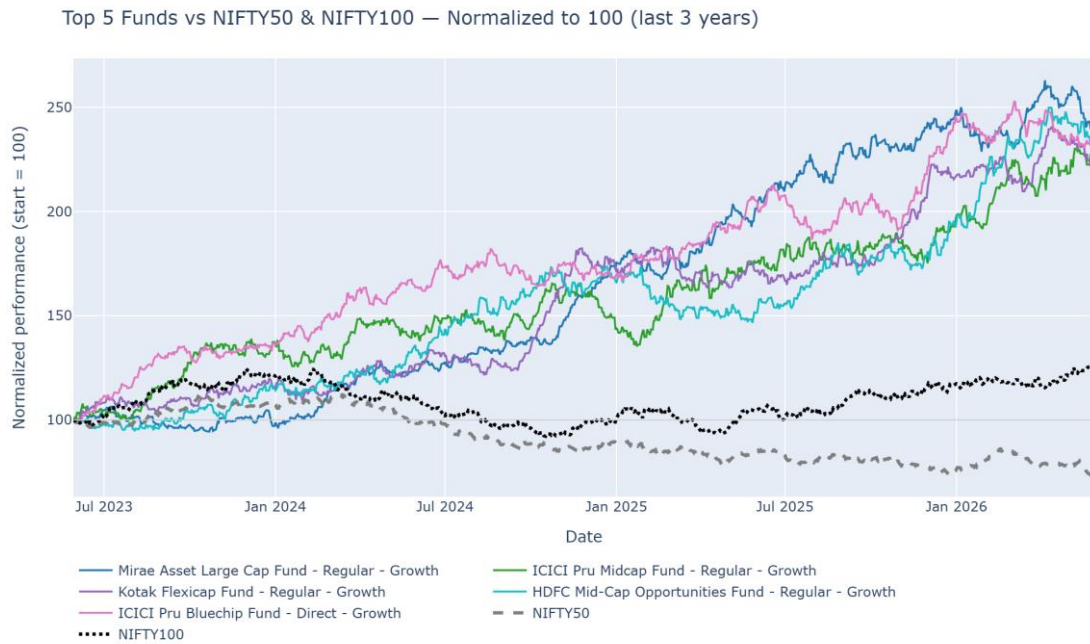


Figure 7.2: Alpha-Beta Performance Comparison Matrix against Benchmarks

## 7.2 Advanced Risk Analytics (VaR, CVaR, HHI)

- **Value at Risk (VaR):** Applied a 95% historical confidence interval to determine the maximum expected daily loss.
- **Sector Concentration:** Utilized the Herfindahl-Hirschman Index (HHI) to flag funds over-exposed to specific sectors (e.g., Banking or IT). A higher HHI indicates a lack of diversification, increasing localized risk.
- **Rolling Metrics:** Visualized the 90-day Rolling Sharpe ratio to track the consistency of fund managers over time.

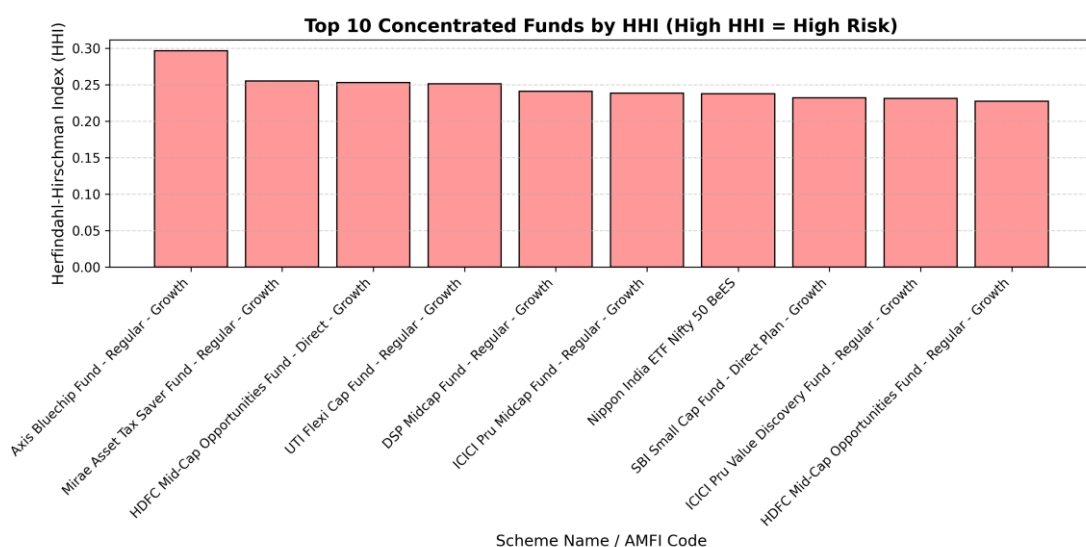


Figure 7.3: Portfolio Structural Risk and Sector Concentration (HHI)

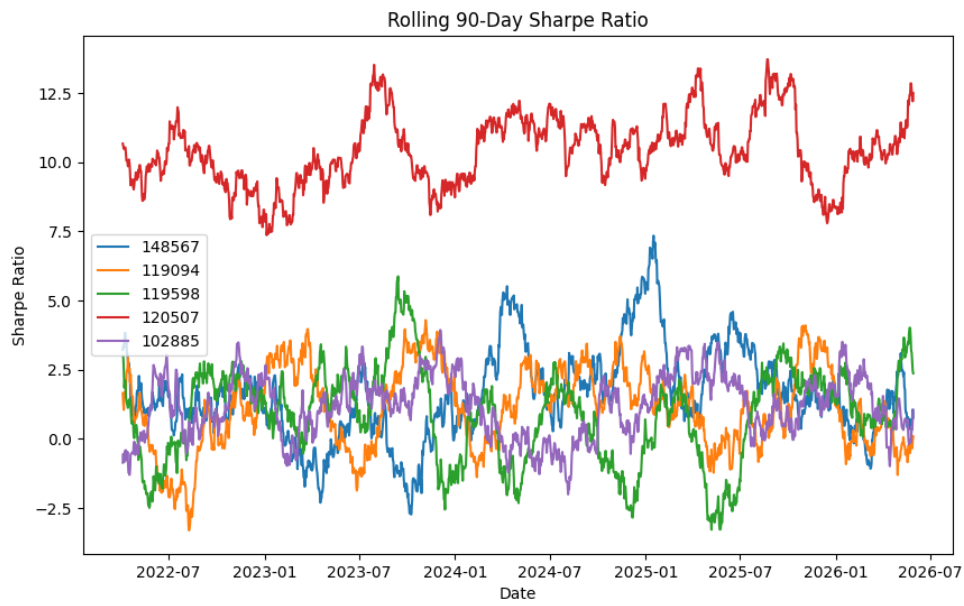


Figure 7.4: 90-Day Rolling Sharpe Ratio Consistency Evaluation

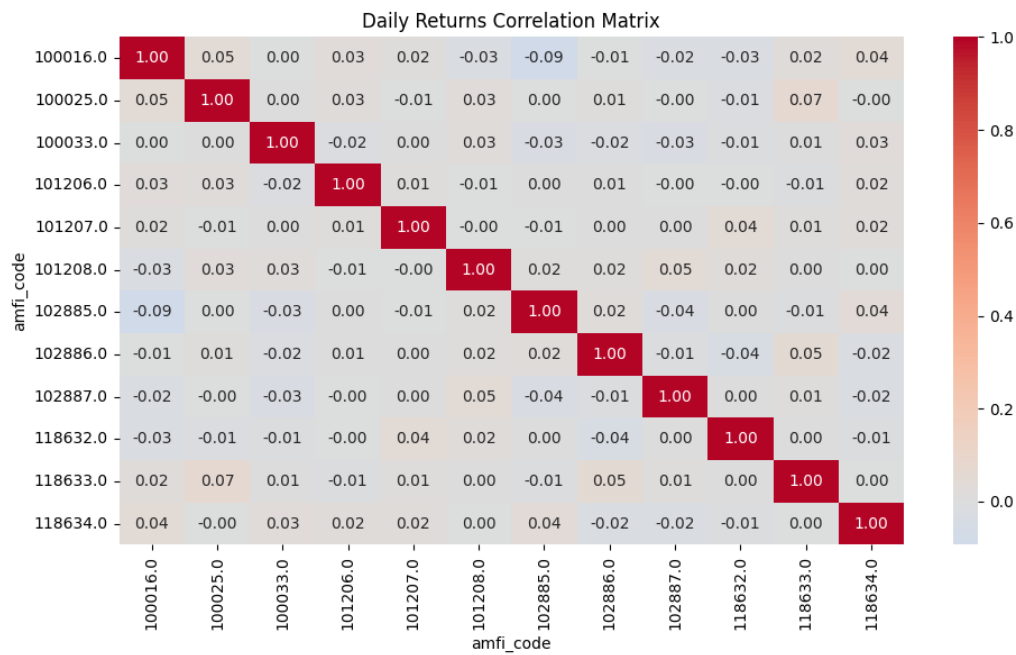


Figure 7.5: Return Inter-dependency & Cross-Correlation Heatmap

## 8. Dashboard Layout & User Guide

To democratize these insights, a comprehensive Power BI dashboard was deployed.

### 8.1 Dashboard Navigation

7. **Industry Overview (Page 1):** Tracks macro KPIs, Folio counts, and AMC dominance.
8. **Fund Performance (Page 2):** Features a Risk vs. Reward scatter plot and the sortable Fund Scorecard Matrix.
9. **Investor Analytics (Page 3):** Visualizes demographic breakdowns, SIP vs. Lumpsum ratios, and state-wise contributions.
10. **Market Trends (Page 4):** Dual-axis charts correlating Nifty 50 movements with retail SIP inflows.

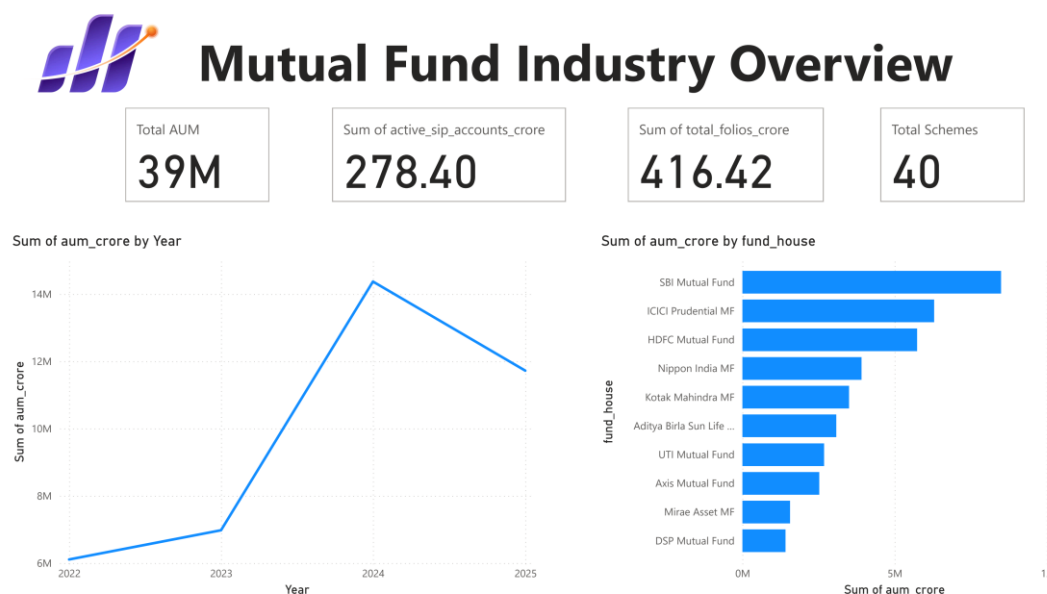


Figure 8.1: Power BI Interface – Executive Industry KPI Dashboard



## Fund Performance Analysis

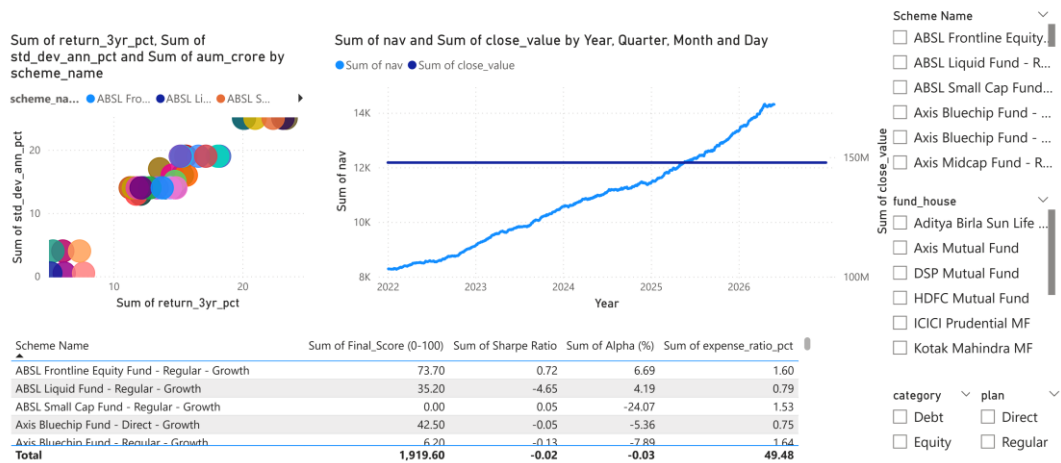


Figure 8.2: Power BI Interface – Fund Performance & Risk-Return Scatter Plot



## Investor Behavior Analysis

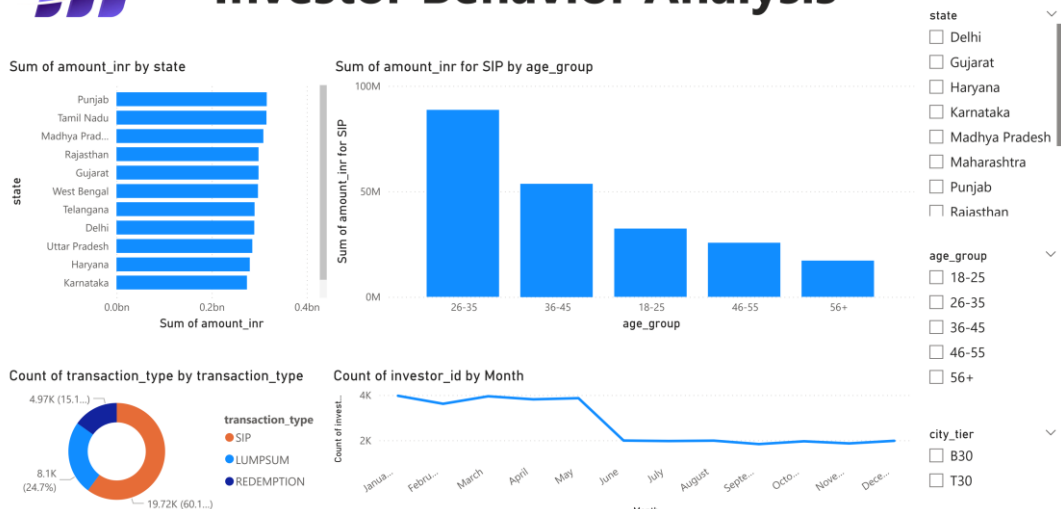
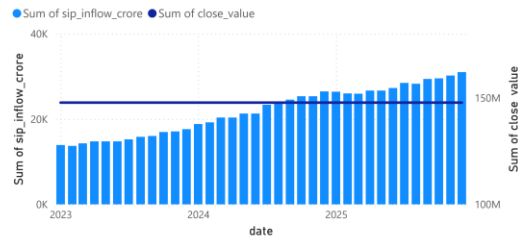


Figure 8.3: Power BI Interface – Investor Demographics and Behavioral Insights

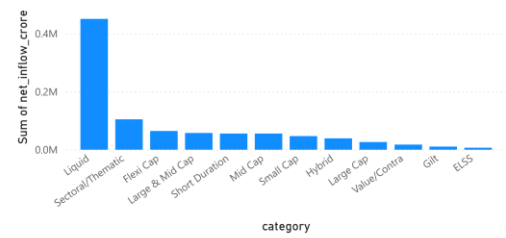


# SIP & Market Trend Analysis

Sum of sip\_inflow\_crore and Sum of close\_value by date



Sum of net\_inflow\_crore by category



| category        | January | February | March | April | May   | June  | July  | August | September | October | November | December | Total  |
|-----------------|---------|----------|-------|-------|-------|-------|-------|--------|-----------|---------|----------|----------|--------|
| ELSS            | 516     | 437      | 500   | 466   | 553   | 472   | 471   | 499    | 537       | 537     | 571      | 521      | 6080   |
| Flexi Cap       | 5603    | 6068     | 4767  | 4947  | 5529  | 4478  | 4869  | 5562   | 5397      | 6004    | 6111     | 4654     | 63989  |
| Gilt            | 744     | 942      | 956   | 784   | 836   | 864   | 959   | 952    | 925       | 898     | 704      | 831      | 10395  |
| Hybrid          | 2967    | 3360     | 2830  | 2955  | 3487  | 3163  | 3291  | 3684   | 3015      | 3314    | 3264     | 3538     | 38868  |
| Large & Mid Cap | 4816    | 5524     | 4243  | 4214  | 4368  | 4610  | 5023  | 5411   | 4528      | 4581    | 5556     | 4878     | 57752  |
| Large Cap       | 2025    | 1925     | 2234  | 2413  | 2076  | 2519  | 2574  | 1940   | 1879      | 2255    | 1870     | 1923     | 25633  |
| Liquid          | 33892   | 32374    | 38681 | 37537 | 41872 | 40486 | 34643 | 41952  | 35308     | 39091   | 40506    | 34933    | 451275 |
| Mid Cap         | 4316    | 4819     | 5061  | 3897  | 5300  | 5047  | 4548  | 3899   | 4960      | 4106    | 4336     | 5023     | 55312  |
| Total           | 73417   | 73631    | 78573 | 74526 | 82661 | 80911 | 75608 | 81601  | 75865     | 79180   | 80323    | 75943    | 932239 |

Figure 8.4: Power BI Interface – Market Trends and Macro Correlations

## 9. Strategic Recommendations for Bluestock Fintech

Based on the data engineering pipeline, analytical models, and business insights generated during this project, the following strategies are recommended to enhance customer engagement and drive business growth:

11. **Deploy Personalized Investment Recommendations:** Utilize the calculated risk-adjusted metrics (Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Fund Score) to build a robo-advisory system that recommends mutual fund portfolios aligned with an investor's risk profile and investment goals.
12. **Strengthen SIP Retention Strategies:** Identify investors with irregular SIP contributions or payment gaps and implement automated reminders, personalized notifications, and relationship manager interventions to improve customer retention and reduce SIP discontinuation.
13. **Expand Market Penetration in B30 Cities:** The geographic analysis indicates significant growth potential beyond Tier-1 cities. Targeted awareness campaigns and digital investment solutions can help increase mutual fund adoption in B30 regions.
14. **Implement Real-Time Risk Monitoring:** Introduce automated alerts for significant portfolio drawdowns, excessive sector concentration, or abnormal market volatility, enabling investors to rebalance their portfolios proactively.
15. **Enable Automated Dashboard Refresh:** Integrate scheduled API-based data ingestion to provide near real-time NAV updates and ensure that business dashboards always reflect the latest market information.

## 10. Project Limitations & Future Scope

### 10.1 Limitations

While the Mutual Fund Analytics Platform provides valuable insights into fund performance and investor behavior, it has a few limitations:

- **Limited Historical Coverage:** The available NAV history spans approximately 4.4 years (2022–2026). As a result, 5-year CAGR and other long-term performance metrics could not be computed reliably.
- **Point-in-Time Performance Metrics:** Risk measures such as Sharpe Ratio, Sortino Ratio, Alpha, and Beta represent a single reporting period and may vary under different market conditions.
- **Historical Risk Assumptions:** The Value at Risk (VaR) model is based on historical return distributions and may not accurately capture rare or extreme market events (Black Swan events).
- **Synthetic Investor Data:** Parts of the investor transaction dataset are simulated for educational purposes and may not fully represent real-world investor behavior during periods of market volatility.
- **External API Constraints:** Live NAV updates depend on third-party APIs, which are subject to availability, rate limits, and periodic service interruptions.

### 10.2 Future Scope

The platform can be further enhanced by incorporating advanced analytics and intelligent decision-support capabilities:

- **Machine Learning Forecasting:** Implement predictive models such as LSTM, ARIMA, or Prophet to forecast short-term NAV movements and market trends.
- **AI-Based Fund Recommendation Engine:** Develop a personalized recommendation system that suggests mutual funds based on an investor's risk profile, investment horizon, and financial goals.
- **Sentiment Analysis:** Integrate NLP techniques to analyze financial news and social media sentiment, enabling the platform to incorporate market sentiment into fund evaluation.
- **Cloud Deployment & Real-Time Analytics:** Deploy the solution on cloud platforms with automated ETL scheduling and real-time dashboard updates for continuous monitoring.
- **Portfolio Optimization:** Extend the platform with portfolio optimization algorithms to recommend diversified investment portfolios based on modern portfolio theory and investor preferences.

## 11. Conclusion

**Mutual Fund Analytics Platform** successfully demonstrates the application of data engineering, financial analytics, and business intelligence to solve real-world investment analysis challenges. By integrating data from multiple sources through an automated ETL pipeline, storing it in a structured SQL database, and visualizing key insights using Power BI, the project provides a comprehensive framework for analyzing mutual fund performance and investor behavior.

The analytical models and dashboards enable users to evaluate fund performance using both return-based and risk-adjusted metrics, identify investment trends, and gain actionable business insights. The project also highlights the importance of data-driven decision-making in the financial services industry and showcases practical skills in Python, SQL, data visualization, and financial analytics.

Overall, the platform serves as a scalable foundation for future enhancements, including machine learning-based forecasting, personalized investment recommendations, and real-time financial analytics.

## 12. References

The following resources were used during the development of this project:

16. [Association of Mutual Funds in India \(AMFI\)](https://www.amfiindia.com/) – <https://www.amfiindia.com/>
17. [MFAPI – Mutual Fund API](https://www.mfapi.in/) – <https://www.mfapi.in/>
18. [Microsoft Power BI Documentation](https://learn.microsoft.com/power-bi/) – <https://learn.microsoft.com/power-bi/>
19. [Python Software Foundation](https://www.python.org/) – <https://www.python.org/>
20. [Pandas Documentation](https://pandas.pydata.org/docs/) – <https://pandas.pydata.org/docs/>
21. [SQLAlchemy Documentation](https://docs.sqlalchemy.org/) – <https://docs.sqlalchemy.org/>
22. [SQLite Documentation](https://www.sqlite.org/docs.html) – <https://www.sqlite.org/docs.html>
23. [Matplotlib Documentation](https://matplotlib.org/stable/) – <https://matplotlib.org/stable/>
24. [Seaborn Documentation](https://seaborn.pydata.org/) – <https://seaborn.pydata.org/>
25. [Investopedia – Financial Metrics and Mutual Fund Concepts](https://www.investopedia.com/) – <https://www.investopedia.com/>

## 13. Appendix

The appendix contains supplementary materials used during the project development and analysis.

### Appendix A: Database Schema

- Star Schema design showing relationships between fact and dimension tables.

### Appendix B: ETL Workflow

- Data extraction, transformation, validation, and loading process.

### Appendix C: Power BI Dashboards

- Dashboard 1 – Industry Overview
- Dashboard 2 – Fund Performance Analysis
- Dashboard 3 – Investor Analytics
- Dashboard 4 – Market Trends & SIP Analysis

### Appendix D: Performance Metrics

- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Value at Risk (VaR)

### Appendix E: SQL Queries

- Sample analytical SQL queries used for KPI generation and reporting.

### Appendix F: Technology Stack

- **Programming Language:** Python
- **Database:** SQLite
- **Libraries:** Pandas, NumPy, SQLAlchemy, Matplotlib, Seaborn, Plotly
- **Visualization:** Microsoft Power BI
- **Version Control:** Git & GitHub

### Appendix G: Abbreviations

| Abbreviation | Description                          |
|--------------|--------------------------------------|
| <b>AMC</b>   | Asset Management Company             |
| <b>AMFI</b>  | Association of Mutual Funds in India |
| <b>AUM</b>   | Assets Under Management              |
| <b>SIP</b>   | Systematic Investment Plan           |
| <b>NAV</b>   | Net Asset Value                      |
| <b>CAGR</b>  | Compound Annual Growth Rate          |
| <b>VaR</b>   | Value at Risk                        |
| <b>HHI</b>   | Herfindahl–Hirschman Index           |
| <b>KPI</b>   | Key Performance Indicator            |
| <b>ETL</b>   | Extract, Transform, Load             |
| <b>OLAP</b>  | Online Analytical Processing         |